Jagan K K

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Summary

Computer Science undergraduate (B.Tech, May 2026) experienced in full-stack web development, AI/ML, and augmented reality game development. Hands-on with React, TypeScript, Node.js, and Python. Built Retrieval-Augmented Generation (RAG) pipelines with LangChain, Google Gemini, and ChromaDB; full-stack applications on TanStack Start, Supabase, FastAPI, and MongoDB; and real-time computer vision systems with OpenCV, YOLOv8, and MediaPipe. Deployed applications on Vercel and Docker. Seeking a software engineering role where I can contribute to AI and full-stack product teams.

Technical Skills

- Languages: Python, TypeScript, JavaScript, C, Java, SQL, HTML, CSS
- Frameworks: React, Node.js, TanStack Start/Router, Vite, Tailwind CSS v4, FastAPI, Flask, LangChain, LangGraph, Vercel AI SDK, Hono, Bun
- AI / ML: RAG, Vectorless RAG, ChromaDB, Google Gemini, Groq, Tavily, OpenRouter, OpenCV, YOLOv8, MediaPipe, deepagents
- Databases: MongoDB, PostgreSQL, Supabase, MySQL, Prisma
- DevOps & Tools: Git, Docker, Vercel, Render, Unity, AR Foundation, IndexedDB

Education

- St. Thomas College of Engineering and Technology, Kannur, Kerala - B.Tech in Computer Science (Sep 2022 - May 2026 Expected). CGPA: 8.38/10.
- Indira Gandhi Public School, Kannur, Kerala - Higher Secondary Education (2019 - 2021). Plus Two: 79%, SSLC: 78%.

Internship

DevOps Intern - Kerala Vision Broadband Ltd. Designed and developed a secure, automated system for backing up network device configurations, with role-based access control and scheduled backups. Implemented a configuration comparator, device-specific command mapping, and WhatsApp notifications on backup status.

Featured Projects

Cerbi - Retro OS Desktop Experience - TypeScript, React, TanStack Start, Supabase, Tailwind CSS v4, Vercel, PostgreSQL

Full-stack web application that recreates a desktop operating system as a retro game-style experience. Built with TanStack Start (React + Vite + Nitro), Supabase, and Tailwind CSS v4, deployed on Vercel.

- Pixel-art SNES-style window chrome with draggable, stackable windows and z-order management.
- News Tome: monthly calendar of news articles with AI-generated summaries via OpenRouter and source attribution.
- Chat Tavern: code-only chat rooms with real-time messaging via Supabase Realtime.
- Daily Trial: 10-question daily quiz for your occupation, with AI-generated questions from GNews articles, score tracking and streak rewards.
- Shop: spend points on cosmetics and items, with buy/equip/unequip item management.
- Music Player (premium): Spotify login, playlist browse/search, Web Playback SDK streaming, and community playlist showcase.

Zithr - Live Streamer & Browser Caching - Python, FastAPI, MongoDB, React, Vite, Docker, IndexedDB

A music streaming platform: import Spotify playlists via CSV, resolve each track to a free stream (Internet Archive), cache audio in the browser via IndexedDB, and expose playlists and streams to other apps through a key-protected public API.

- Backend: FastAPI + MongoDB (Motor), Dockerized, with CORS and API key auth (header/Bearer/query).
- Frontend: React + Vite with a player bar, sidebar, and API key manager.
- Browser caching layer in IndexedDB for offline audio.

Stateful RAG Pipeline with Vector Deduplication - Python, LangChain, Google Gemini, ChromaDB, SQLRecordManager, deepagents

An intelligent Retrieval-Augmented Generation system built with LangChain, Google Gemini, and ChromaDB that answers questions from PDF documents.

- Stateful incremental indexing using SQLRecordManager with sha256 hashing - unchanged chunks are skipped, eliminating duplicate embedding API costs and reducing runtime.
- Google Gemini embeddings (gemini-embedding-001) and gemini-1.5-flash for high-accuracy reasoning.
- Strict QA guardrails preventing hallucinations by restricting answers to retrieved document context.
- Autonomous deep agent that dynamically retrieves relevant chunks before answering.

KL-CODE - Terminal AI Coding Assistant - TypeScript, Bun, Hono, React, Prisma, PostgreSQL (Neon), Vercel AI SDK, OpenRouter

An interactive terminal (TUI) application that chats with AI language models via OpenRouter, OpenAI, or local Ollama to help with software development tasks.

- Monorepo with Bun and npm workspaces: CLI (React TUI), server (Hono API with SSE streaming), database (Prisma + PostgreSQL via Neon), and shared package.
- Tool execution: readFile, writeFile, bash, grep, glob, webfetch.
- Agent modes: plan, build, debug, review.

AI AR Data Augmentation (Memory Palace) - Python, Flask, Unity, AR Foundation, OpenCV, MiDaS, MongoDB, YOLO

An AI + Augmented Reality system that transforms uploaded PDF documents into interactive 3D AR cards, visualized in AR on mobile via Unity and AR Foundation.

- Admin web portal (Flask) and user mobile app that deliver documents without requiring login.
- Geofencing plus YOLO-based object detection so only users at specified locations can access data.
- AI depth estimation (MiDaS) and document summarization; UV mapping of images to 3D cards.

SketchPlay AI - Python, OpenCV, YOLOv8, MediaPipe, RNN, AR

A real-time 2D augmented game where users draw simple objects on paper; a custom-trained AI model recognizes the sketch (maze, grid, square) and turns it into a 2D game platform rendered into the user's environment via the device camera.

- Collaborated with a team to build the real-time AR game pipeline.
- Trained and implemented deep learning models including YOLOv8 and an RNN for object detection and interaction.
- Optimized to run at 30 FPS on medium-spec laptops, with gameplay controlled via hand gestures using MediaPipe.

Languages

English, Hindi, Malayalam